

ARSHIYA KHAN

SUMMARY

Ph.D. candidate specializing in AI-driven cybersecurity and large-scale threat detection systems. Over 6 years of experience in designing and deploying machine learning pipelines to detect anomalies, vulnerabilities, and adversarial behaviors.

EDUCATION

Ph.D. Electrical and Computer Engineering

2019 - 2026 (*expected*)

University of Delaware

Newark, DE

- Advisors: Dr. Xing Gao, Dr. Chase Cotton
- Dissertation: Red-teaming LLMs: Bypassing security guardrails of AI-driven systems by hiding malicious intent in prompts. Converted unsafe prompt to safe by token manipulation.

M.S. Cybersecurity

2017 - 2019

University of Delaware

Newark, DE

- Advisor: Dr. Chase Cotton
- Thesis: Taxonomy of network traffic to perform ML-based anomaly detection for new attack signals

TECHNICAL SKILLS

Machine Learning: LLM safety and alignment, Malware detection, Vulnerability repair, Statistical analyses, Natural language processing (NLP), Reinforcement learning

Network Security: Packet analysis, Network traffic analysis, Pen testing, Reverse engineering, System hardening, Fuzzing, Cryptography

ML Frameworks & Tools: Torch, TensorFlow, Scikit-learn, Keras, Hugging Face, NumPy, Pandas, Scipy, OpenCV, PIL

Programming Languages: Python, R, MATLAB, Bash, Java, C++, MySQL

PUBLICATIONS

1. **Code Vulnerability Repair with LLM using Context-Aware Prompt Tuning**
IEEE Security and Privacy Workshops (SPW) 2025
Improved buffer overflow vulnerability repair by 74% on GitHub Copilot. Used prompt tuning to introduce security and code knowledge to code-generating LLMs.
2. **Ensemble Defense System: A Hybrid IDS Approach for Effective Cyber Threat Detection**
IEEE International Telecommunication Networks and Applications Conference 2023
Developed a hybrid intrusion detection model by integrating signature-based and anomaly-based IDS tools. Designed an interactive tracker for tracking and visualizing threat vectors.
3. **Efficient attack detection in IoT devices using feature engineering-less machine learning**
International Journal of Computer Science and Information Technology 2022
Designed a lightweight malware detection model that removed the feature-engineering process from the ML pipeline. Deployable on battery-powered, low-memory IoT devices.
4. **Detecting Attacks on IoT Devices using Featureless 1D-CNN**
IEEE International Conference on Cyber Security and Resilience (CSR) 2021
Used pcap files as raw time-series signals to train a 1D-CNN for malware detection. Emphasize the role of headers in malware detection.
5. **A Feature Taxonomy for Network Traffic**
University of Delaware, Department of Electrical and Computer Engineering 2019
Developed a taxonomy model to detect anomalous activities detected in packets, flows, and sessions. Performed extensive analysis of ML-based anomaly detection techniques based on raw, behavioral, and temporal features. Highlighted anomalies detected by network traffic scanners such as Zeek and Wireshark.

PROJECTS	1. JPMC Data for Good Hackathon 2021 Led a team of 4 to design a forecasting model with Linear Regression with a focus on sparse data handling.	
	2. Twitter Bot Detection and Classification 2019 Built a Multinomial Naive Bayes model, achieving 91% accuracy on detecting harmful bot accounts on Twitter and classifying them into political, social, and personal harm categories.	
WORK EXPERIENCE	Graduate Teaching Assistant University of Delaware - Developed and taught a graduate-level class, “Topics in Cybersecurity”, on implementing supervised ML techniques for anomaly detection in large-scale data sources like packet captures or compiled network security logs from Zeek. - Collaborated with professors to guide undergraduate, graduate, and mid-career professionals. Teaching Assistant for CPEG694 System Hardening and Protection, TOPICS19F Topics in Cybersecurity, CPEG665 Introduction to Cybersecurity, CPEG697 Advanced Cybersecurity. - Received recognition for creating a supportive and inclusive environment in the classroom as a TA.	2019 – Present
	Backend & Analytics Developer Women Plus Blockchain - Built 50 web crawlers to collect data from blockchain news outlets; implemented a scalable pipeline on AWS console to filter relevant articles using SEO-based screening - Boosted subscriber growth by 17% by automating user engagement metrics on Google Analytics dashboard in 4 months	2018.06 - 2018.08
	Systems Engineer Tata Consultancy Services Ltd - Led an application team of 5 members to support and maintain a telecom banking application, serving daily transactions - Deployed changes to the production environment after application testing, resolved critical incidents, maintaining 99% uptime SLA - Cut reporting costs by 90% by automating daily business intelligence report generation using VBA	2014.09 - 2017.04
TEAMWORK EXPERIENCE	• Led peer mentoring organization EmPOWER at University of Delaware	2021-2023
	• Associate, Hispanic Latino Graduate Student Association	2022
	• Treasurer, Indian Graduate Student Association	2021
	• UMBC CyberDawgs CTF at the University of Maryland	2019